

Homework 1: Discrete State Dynamic Programming

ECONS 509, Fall 2026

1 Overview

I solve the two-state Huggett household problem at the fixed prices implied by $K^0 = 5$. The final numerical specification uses modified Howard improvement, sparse CSR power iteration, an exp-log asset grid on $[0, 5]$, and $N = 5000$. The Bellman formulation and Euler derivation are given in `manual/derivation.pdf`; generated tables and figures are included from `results/`.

2 Solver Comparison: Part (a)

The baseline uses a uniform $[0, 20]$ grid with $N = 1000$ and zero initial values. Convergence requires

$$d(x) = \max_m \frac{|T(x)_m - x_m|}{|x_m| + 1} \leq 10^{-8}.$$

Pass counts include the terminating check, and solver times exclude the common utility-matrix construction.

Table 1: Baseline solver comparison, $N = 1000$ on $[0, 20]$.

Method	Passes	Updates	Inner steps	Seconds	Exit metric	Status
vfi	379	378	0	0.923665	9.71806129077e-09	converged
howard	15	14	0	0.063859	6.01925131926e-16	converged
modified_howard	19	18	360	0.055136	9.33987779834e-09	converged
gradient	20000	20000	0	51.552503	1.25760126001	capped_unconverged
adam	20000	20000	0	51.727502	0.125113107262	capped_unconverged

VFI, Howard, and modified Howard converge to identical policies in 379, 15, and 19 passes. Modified Howard is the fastest converged method at 0.055136 seconds, so I select it. The contraction benchmark $\log(10^{-8})/\log(0.96) = 451.244$ is larger than the observed VFI count because it is a contraction-based error-reduction benchmark, not the exact stopping count for the relative Bellman metric.

The fixed gradient step $\eta = 0.0005$ was preregistered as a conservative choice: at the baseline size, the bound on $\|I - \beta Q\|_2^2$ is below 2000, so the step uses the corresponding inverse scale. Fixed-step gradient descent and Adam both hit the 20,000-pass cap and remain unconverged. The requested benchmark $(1 - \beta)^{-2} = 625$ is reported separately from the measured Euclidean condition numbers. In this problem, the measured $\text{cond}_2(J^T J)$ is much larger, so 625 should not be read as the measured 2-norm condition number of this nonnormal Jacobian.

Table 2: Contraction and conditioning diagnostics, $J = I - \beta Q_G$.

Quantity	Value
Contraction expression	451.24401589
Observed VFI outer passes	379
Specified $1/(1-\beta)^2$ expression	625
zero_values: sigma_min	0.00131755850852
zero_values: sigma_max	30.3591830961
zero_values: condition_J	23041.9999566
zero_values: condition_JtJ	530933762
howard_solution: sigma_min	0.0199827991366
howard_solution: sigma_max	2.03095066516
howard_solution: condition_J	101.634943697
howard_solution: condition_JtJ	10329.6617804

3 Invariant Distribution: Parts (b)–(d)

Using the converged Howard policy, I compare power iteration, the eigenvector method, and direct solution of $Q_G^T \pi = \pi$ with both dense and CSR representations.

Table 3: Invariant-distribution methods for the common Howard policy.

Method	Seconds	Power updates	Raw minimum	Correction mass	Stationarity
dense_power	0.238683	212	0	-0	7.76899378163e-13
dense_eigen	0.074303	n/a	-8.06313061579e-28	4.45487966523e-26	1.73472347598e-17
dense_direct	0.036680	n/a	0	-0	3.1918911958e-16
sparse_power	0.003668	212	0	-0	7.76906317057e-13
sparse_eigen	0.005532	n/a	-4.0315653079e-28	2.01578265395e-26	1.38777878078e-17
sparse_direct	0.005262	n/a	0	-0	8.48773347979e-18

Table 4: Sup-norm agreement among invariant distributions.

Comparison	Maximum difference	Pair
dense	5.49010836792e-12	dense_power, dense_direct
sparse	5.48994877336e-12	sparse_power, sparse_eigen
across_representations	5.49011530682e-12	dense_direct, sparse_power
all_six	5.49011530682e-12	dense_direct, sparse_power

All six solutions pass the registered checks, with maximum disagreement about 5.49×10^{-12} . Sparse power iteration is the fastest measured sparse method at 0.003668 seconds, so I select power iteration with CSR for the remaining experiments.

4 Asset Range: Part (e)

At $N = 1000$, I test uniform grids with $k_{\max} = 1, 2, 5, 10, 20, 40$.

The ranges ending at 1 and 2 are rejected because they place positive mass at the top node. I select $k_{\max} = 5$, the smallest tested range with zero top-node mass and an upper support endpoint well below the grid boundary.

Table 5: Uniform asset-range comparison at $N = 1000$.

k_max	Top mass	Support endpoint	Support share	Compute seconds	Range status
1	0.0237509414545	1	1.000000	0.085518	rejected_truncated
2	6.01277209443e-05	2	1.000000	0.082441	rejected_truncated
5	0	3.21821821822	0.634000	0.074443	eligible
10	0	3.21321321321	0.322000	0.072260	eligible
20	0	3.26326326326	0.164000	0.072538	eligible
40	0	3.24324324324	0.082000	0.074047	eligible

5 Grid Construction: Part (f)

At $N = 1000$ on $[0, 5]$, I compare the uniform grid with

$$k_n = \exp\left[\frac{n-1}{N-1} \log(1 + k_{\max})\right] - 1.$$

Table 6: Uniform and exp-log grids, $N = 1000$ on $[0, 5]$.

Quantity	Uniform	Exp-log
Outer passes	19	19
Updates	18	18
Inner steps	360	360
Exit metric	8.07103857078e-09	8.0834666578e-09
Solver seconds	0.0566389579981	0.0530802919966
Power updates	172	168
Distribution seconds	0.00352920899604	0.00297812499775
Compute seconds	0.0831304579915	0.0757789579875
Maximum grid step	0.00500500500501	0.0107516734031
Mean assets	0.450551165217	0.450853905799
Top mass	0	0
Support endpoint	3.21821821822	3.21407251709
Support share	0.634	0.799
Slack count	1978	1945
Slack mass	0.90977107721	0.911966346902
Upper choices	0	0
Euler maximum	0.00412096375367	0.00612421604424
Euler mean	0.00120403136135	0.00108912237111
Conditional weighted mean	0.00147199905766	0.0008046570183
Weighted maximum: max(qE)	0.000102998957978	6.13602452215e-05
Supported slack maximum	0.00412096375367	0.00407219098769

Both grids pass all applicable checks. I select the exp-log grid because it has lower mean and distribution-weighted Euler errors, although its unweighted maximum residual is slightly higher.

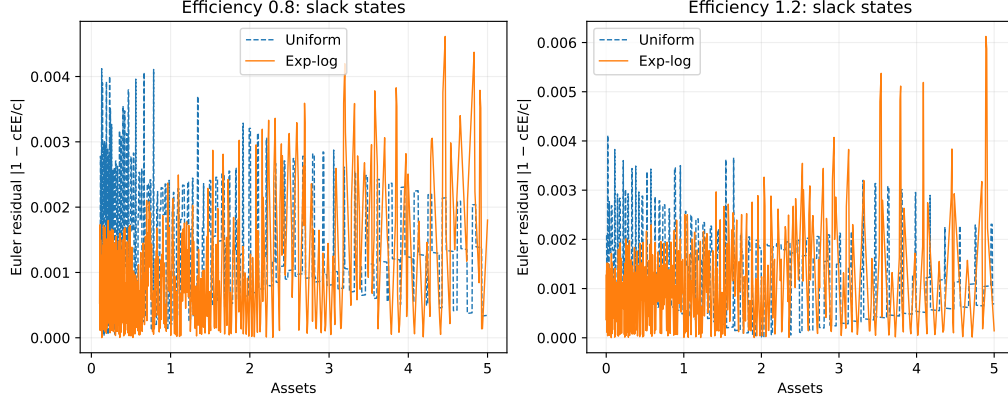


Figure 1: Euler-residual comparison for the uniform and exp-log grids.

6 Grid Size: Part (g)

I compare $N = 100, 500, 1000, 2000, 5000$ under the selected configuration.

Table 7: Grid-size comparison on the exp-log $[0, 5]$ grid.

Quantity	N=100	N=500	N=1000	N=2000	N=5000
Outer passes	19	19	19	19	19
Updates	18	18	18	18	18
Inner steps	360	360	360	360	360
Exit metric	8.03364122753e-09	8.07799018063e-09	8.0834666578e-09	8.07986477462e-09	8.08314692909e-09
Utility seconds	0.000309292001475	0.00679545800085	0.0241041250047	0.112672458003	0.655352125003
Solver seconds	0.00388824900438	0.0156604590011	0.0537732079974	0.212434167006	1.285760583
CSR construction seconds	0.000102291996882	0.000170540995896	0.000141750002513	0.000179458002094	0.000298332997772
Power seconds	0.00271195799724	0.00287229099922	0.00301779100118	0.00373687600222	0.00624258299649
Power updates	168	169	168	168	168
Compute seconds	0.00701179099997	0.0254987489971	0.0810368740058	0.329022959013	1.947653624
Stationarity residual	7.87911402789e-13	7.53008766452e-13	8.48758563432e-13	8.50611248104e-13	8.52012904673e-13
Maximum grid step	0.107614708755	0.0215055689019	0.0107516734031	0.00537555790023	0.00215015611502
Mean assets	0.439732993184	0.45019078318	0.450853905799	0.450965370158	0.451126535212
Top mass	2.67276471009e-53	0	0	0	0
Support endpoint	3.02929208117	3.22015069022	3.21407251709	3.2072684858	3.17018140386
Support share	0.78	0.802	0.799	0.793	0.7888
Slack count	193	972	1945	3891	9730
Slack mass	0.896989880687	0.911024525526	0.911966346902	0.912035439329	0.912256038332
Upper choices	1	0	0	0	0
Euler maximum	0.0469940723466	0.0102519768383	0.00612421604424	0.00293533782662	0.00118455627353
Euler mean	0.0117693950252	0.00225801889311	0.00108912237111	0.000563182483879	0.000223018069011
Conditional weighted mean	0.0087004085652	0.00150289123749	0.0008046570183	0.000395872498082	0.000155907135197
Weighted maximum: $\max(qE)$	0.000767549255638	0.000121869663975	6.13602452215e-05	1.72349708195e-05	6.44444556925e-06
Supported slack maximum	0.0469940723466	0.00780556323862	0.00407219098769	0.00197919372155	0.000966044811544

All five solutions converge in 19 passes. Euler errors fall steadily as N increases, while the $N = 5000$ run remains computationally modest at about 1.95 seconds. I therefore select $N = 5000$ because it gives the lowest tested Euler errors at an acceptable runtime.

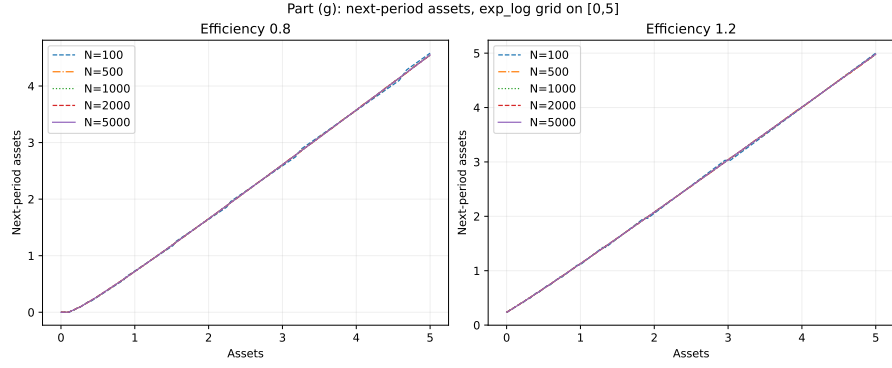


Figure 2: Asset policies across grid sizes.

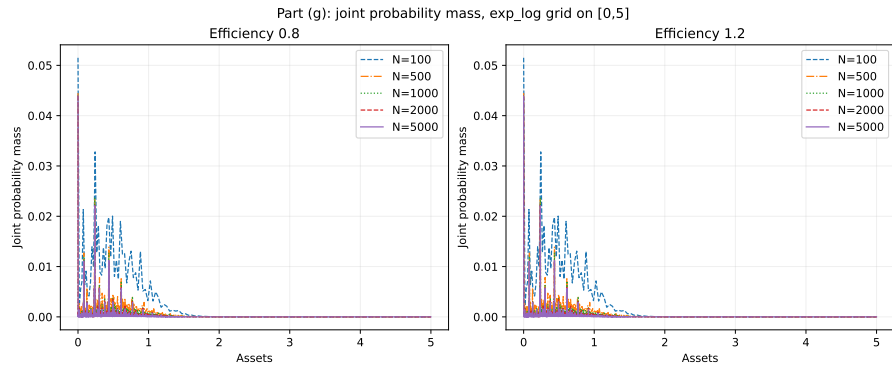


Figure 3: Joint invariant probability masses across grid sizes.

7 Euler Accuracy: Part (h)

For slack borrowing states, the consumption-equivalent residual is

$$c_{n,s}^{EE} = \left[\beta(1+r^0) \sum_{s'} P_{ss'}(c_{n,s,s'}^{\text{next}})^{-\sigma} \right]^{-1/\sigma}, \quad E_{n,s} = \left| 1 - \frac{c_{n,s}^{EE}}{c_{n,s}} \right|.$$

The weighted statistics use probabilities normalized over the slack-state set.

Table 8: Euler residuals at the selected $N = 5000$ grid.

Quantity	Value
slack_count	9730
slack_mass	0.912256038332
upper_choice_count	0
maximum	0.00118455627353
mean	0.000223018069011
weighted_mean	0.000155907135197
weighted_maximum	6.44444556925e-06
supported_maximum	0.000966044811544

The final mean, maximum, and conditional weighted mean residuals are 2.23018×10^{-4} , 1.18456×10^{-3} , and 1.55907×10^{-4} . No selected policy chooses the upper grid bound.

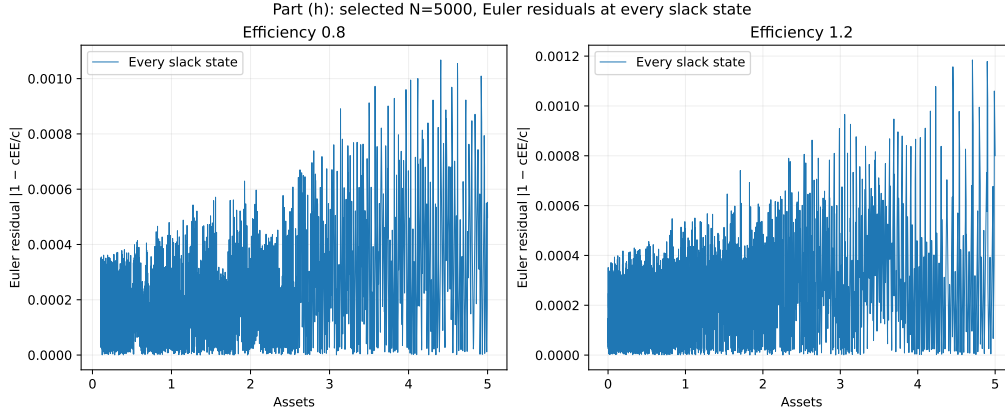


Figure 4: Euler residuals at the selected grid size.

Across the five saved grid sizes, the fitted log-log slopes are 1.013287 for the mean error and 0.933748 for the maximum error.

Table 9: Log-log Euler-error scaling across five grids.

Error summary	Intercept	Slope	Used grids	Reason
mean	-2.19734219457	1.01328727392	5	available
maximum	-0.957782213685	0.933748380727	5	available

These slopes are close to one, which is consistent with approximately first-order error reduction over the tested grid resolutions.

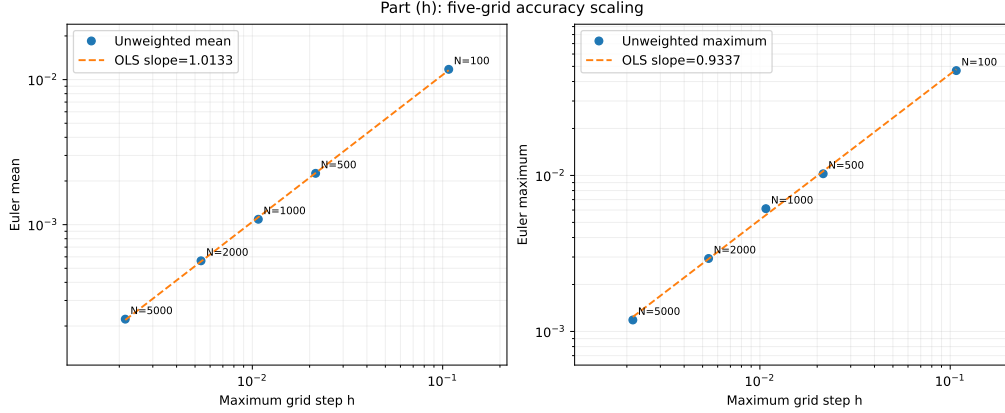


Figure 5: Mean and maximum Euler errors against the maximum grid step.

Final checks. All nine selected-grid checks pass. The full Stage-4 evidence records 202 passed checks and two expected rejected trial ranges. The reproduction check matches 330 saved arrays and 8,514 JSON numerical values with zero numerical difference; the capped gradient and Adam runs remain correctly labeled unconverged.

8 Independent Verification

Manual-kernel check. On the $N = 100$ uniform $[0, 20]$ grid, both the manual and implementation VFI runs require 492 passes. The maximum value-function difference is 3.71×10^{-9} and there are zero policy mismatches, satisfying the 10^{-8} comparison criterion.

Final Euler check. My verification entry in `log.md` records final Euler mean and maximum values of 0.000223018 and 0.001184556, matching the saved Stage-4 results to numerical precision.

These checks are separate from the execution agent’s automated Stage-3 and Stage-4 verification.

9 AI-Use Note

I used Codex (GPT-6; exact build unavailable) through a desktop/local-folder agent workflow to implement the production solvers and tests, run the numerical experiments, and generate the results, tables, figures, and reproduction evidence. I also used ChatGPT text chat to help draft and revise student-owned workflow files, later manual-kernel maintenance edits, and this report draft. I made the recorded runtime choices and reasons, reviewed the evidence, recorded the verification in `log.md`, and am responsible for the final report and Stage-5 commit. The report-check agent may correct only numbers, references, and typographical errors.

10 Conclusion

Modified Howard with CSR power iteration on the $N = 5000$ exp-log $[0, 5]$ grid passes the registered checks and yields small Euler residuals with approximately first-order scaling over the tested resolutions.